Akoizo Original Practice - Form 2 - Answer key

Division B | Rocks And Minerals | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Equal axes and right angles describe the conventional…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. Monoclinic system

B. Orthorhombic system

C. Cubic system

D. Triclinic system

Correct: C

2. (2 points) A two-stage cooling clue is…

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. Metallic luster

B. Perfect sorting

C. Porphyritic texture

D. One uniform streak

Correct: C

3. (2 points) Best reference design includes…

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Teacher names

B. Only specimen prices

C. Confusions and diagnostic distinctions

D. Only one photograph

Correct: C

4. (2 points) Olivine contains…

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. No silicon

B. Isolated tetrahedra

C. Only metal atoms

D. A pure carbonate lattice

Correct: B

5. (2 points) Cementation is…

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Only breaking crystals

B. Binding grains by precipitated material

C. Moving grains downstream

D. Melting a whole rock

Correct: B

6. (2 points) The best identification uses…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. Several diagnostic observations

B. Color alone

C. One nickname

D. Only specimen size

Correct: A

7. (2 points) Ore status depends partly on…

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Economic extraction conditions

B. Color alone

C. Crystal size only

D. One fixed universal value

Correct: A

8. (2 points) Nonfoliated means…

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. No strong planar fabric

B. Never metamorphosed

C. Always igneous

D. No pressure occurred

Correct: A

9. (4 points) Why is slow cooling insufficient to predict size?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

answer: Available space, nucleation, fluid supply, and growth time also matter. (4 points).

10. (4 points) Why separate texture and composition?

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

answer: Different cooling histories can produce different textures from similar compositions. (4 points).

11. (4 points) Design a calcite-quartz comparison.

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

answer: Compare hardness, cleavage/fracture, and supplied carbonate reaction evidence. (4 points).

12. (4 points) Why does crystal elongation not prove chain structure?

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

answer: External habit reflects growth and is not uniquely determined by one connectivity group. (4 points).

13. (4 points) Why is quartz sandstone a rock?

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

answer: It is an aggregate of grains and often cement, even if quartz dominates. (4 points).

14. (4 points) Explain a density calculation.

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

answer: Divide measured mass by specimen volume and report units and porosity limitations. (4 points).

15. (4 points) Why distinguish surface alteration?

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

answer: Weathering can mask the original mineral’s diagnostic properties. (4 points).

16. (4 points) Differentiate mineral and rock cleavage.

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

answer: Mineral cleavage is crystal breakage; rock cleavage is an aggregate planar fabric. (4 points).

17. (4 points) Distinguish symmetry and habit.

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

answer: Symmetry describes structural relationships; habit describes the realized external growth form. (4 points).

18. (4 points) What supports an igneous interpretation?

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

answer: Interlocking primary crystals, appropriate mineral associations, and field relationships. (4 points).